

Two-Sample Test for Laws of Random Probabilities via Optimal Transport



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Bocconi University

The 5th Italian Meeting on Probability and Mathematical Statistics

June 11, 2026

Joint work with



Marta Catalano

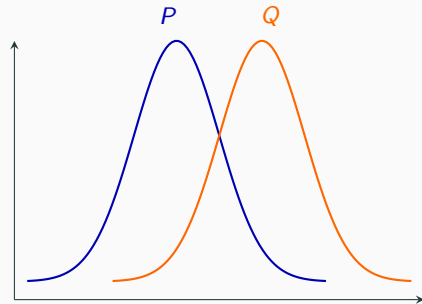
LUISS 
Università di Roma



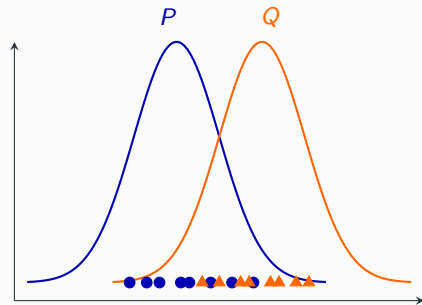
Hugo Lavenant



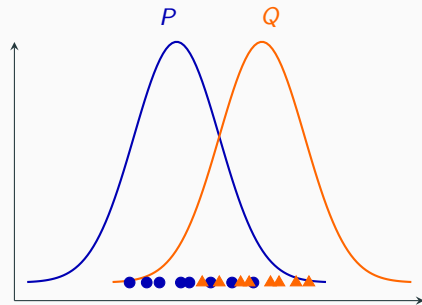
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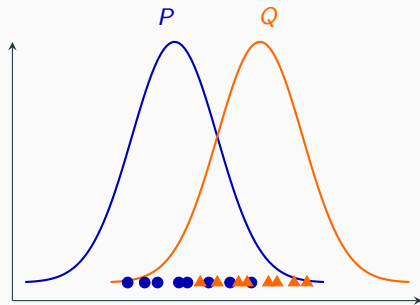
Do they come from the same law?

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(unknown) $P, Q \in \mathcal{P}(\mathbb{R})$

(Observed) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P, \quad Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} Q.$

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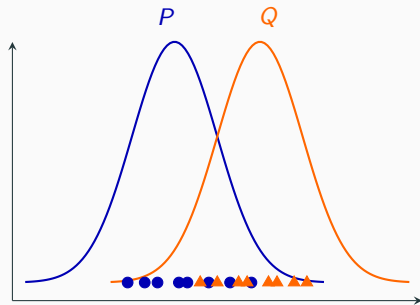
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$$T_n = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - \hat{G}_n(x)|$$



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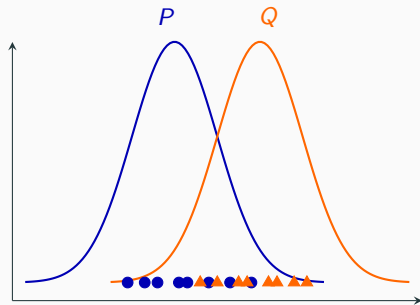
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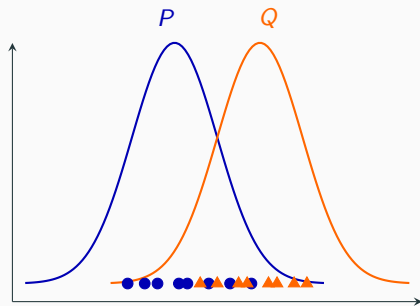
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$$d\left(\frac{1}{n} \sum_{i=1}^n \delta_{X_i}, \frac{1}{n} \sum_{i=1}^n \delta_{Y_i}\right) \text{ is large}$$

Two-Sample Testing in $\mathbb{R}^d$

Works in $\mathbb{R}^d$:

- Gretton et al. (2012) — maximum mean discrepancy
- Wang et al. (2022) — kernel projected Wasserstein
- Hu and Lin (2025) — max-sliced Wasserstein
- $\vdots$

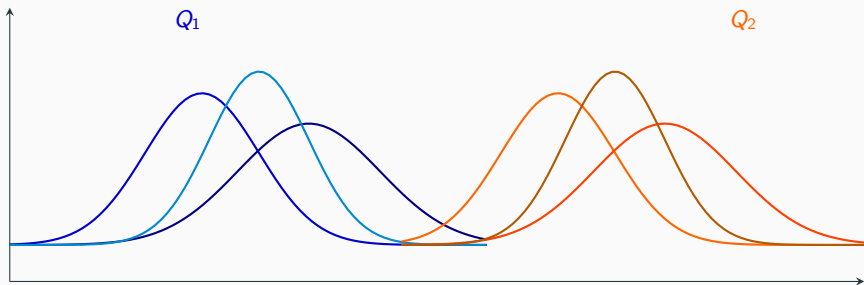
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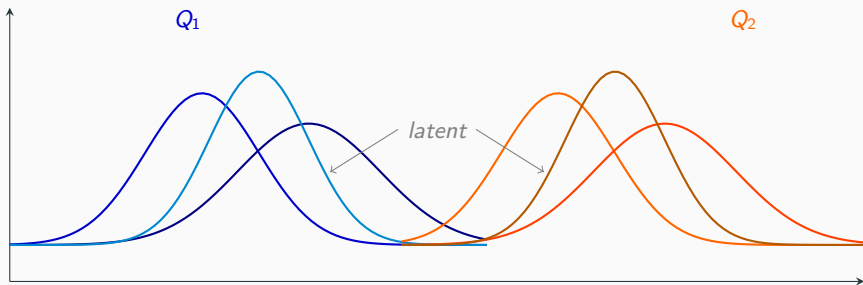
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What about **beyond** $\mathbb{R}^d$? What if the observed samples are themselves **probability measures**?

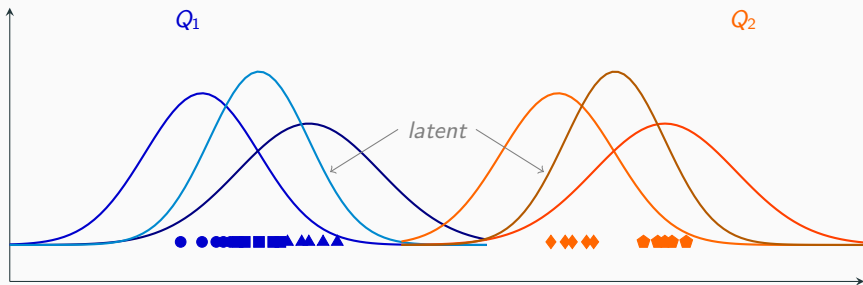
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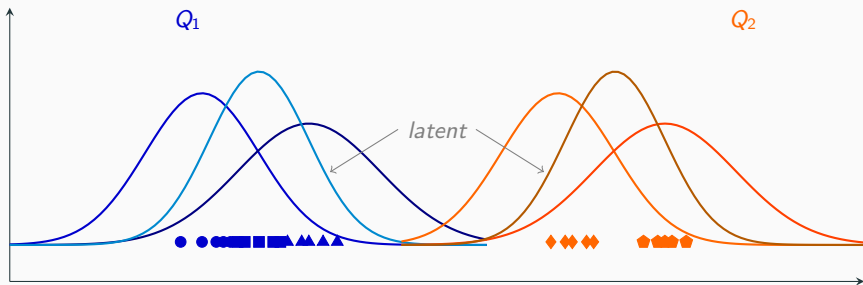
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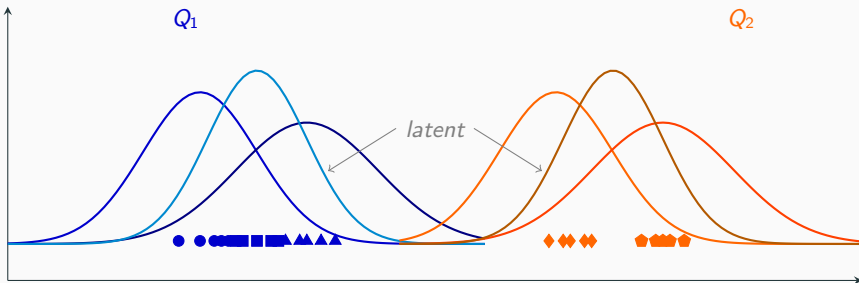


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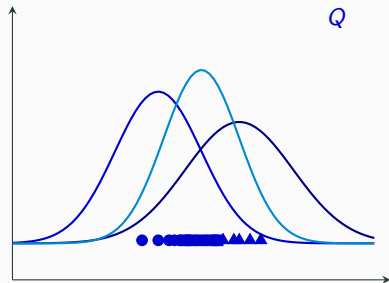
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→ Need to work at the level of **Random Probability Measures**

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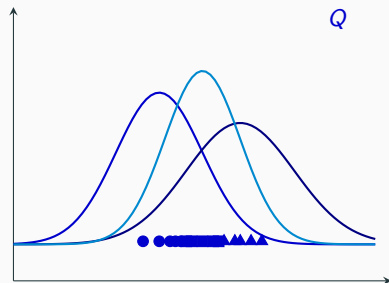
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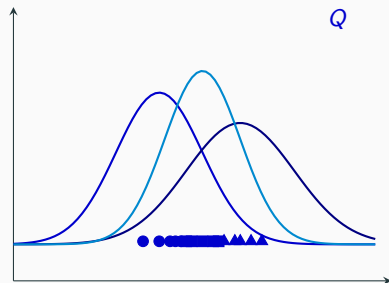
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Notation: $\mathbf{X}_{n,m} \sim Q$



Hierarchical Samples in the Literature

- Baxter (2000) — meta-learning, distribution over tasks
- Szabó et al. (2016) — distribution regression
- Haviv et al. (2024) — generative modelling over distributions
- Warren et al. (2025) — principal curves in Wasserstein space

Limitations of the Hierarchical Sampling Framework

- Each row of $\mathbf{X}_{n,m}$ has the law

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- **But for m large:**

$$Q^1 = Q^2 \iff \mathbf{X}_{n,m} \stackrel{d}{=} \mathbf{Y}_{n,m}.$$

Distance-based Two Sample Test

- **Problem.** $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$; test $H_0 : Q^1 = Q^2$ vs $H_1 : Q^1 \neq Q^2$.

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- **Strategy.** Given hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$, reject H_0 if $d(\mathbf{X}_{n,m}, \mathbf{Y}_{n,m})$ is large.

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- **Ingredients.** A distance d ; a threshold.

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Metrics on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$: Wasserstein over Wasserstein

- $(\mathbb{X}, d_{\mathbb{X}})$ Polish space with bounded diameter.
- **Step 1:** Using $d_{\mathbb{X}}$ on $\mathbb{X}$, define Wasserstein distance $\mathcal{W}$ on $\mathcal{P}(\mathbb{X})$: for $P^1, P^2 \in \mathcal{P}(\mathbb{X})$,

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- **Step 2:** Using $\mathcal{W}$ on $\mathcal{P}(\mathbb{X})$, for $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \inf_{(\tilde{P}^1, \tilde{P}^2) \sim \Gamma(Q^1, Q^2)} \mathbb{E}[\mathcal{W}(\tilde{P}^1, \tilde{P}^2)] .$$

$\mathcal{W}_{\mathcal{W}}$ is called **Wasserstein over Wasserstein (WoW)**.

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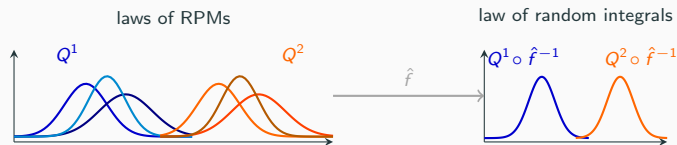
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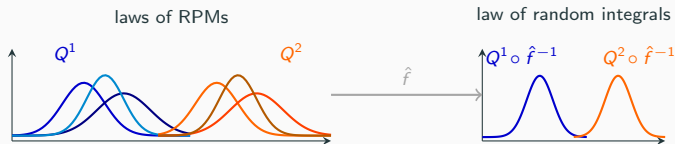
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- **Hierarchical Integral Probability Metric** (Catalano and Lavenant 2024):

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}).$$

- $\mathcal{F} = \text{Lip}_1(\mathbb{X}, \mathbb{R})$: $d_{\mathcal{F}} = d_{\text{Lip}}$.

Properties of $\mathcal{W}_{\mathcal{W}}$ and d_{Lip}

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- $\mathbb{E}[\mathcal{W}_{\mathcal{W}}(\hat{Q}_{n,m}, Q)] \rightarrow 0$ **slower than** any polynomial rate.
- $\mathbb{E}[d_{\text{Lip}}(\hat{Q}_{n,m}, Q)] \rightarrow 0$ at **polynomial rate**.

(Catalano and Lavenant 2024)

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Theorem

Let $\mathcal{G}$, $\mathcal{H}$ be uniformly bounded function classes generating IPMs on $\mathcal{P}(\mathbb{X})$ and $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ respectively, with $\mathcal{H} \subseteq \text{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{I}_{\mathcal{G}})$. Then for all $\theta \in (0, 1)$, $n, m \geq 1$,

$$\mathbb{P}\left(\left|\mathcal{I}_{\mathcal{H}}(\hat{Q}_{n,m}^1, \hat{Q}_{n,m}^2) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\right| \geq r(n, m, \theta)\right) \leq \theta,$$

where

$$r(n, m, \theta) = O\left(\mathcal{R}_n(\mathcal{H}) + \mathcal{R}_m(\mathcal{G}) + \frac{1}{\sqrt{n}}\right).$$

Theoretical Threshold via Rademacher Complexity

$\mathbb{X} \subset \mathbb{R}^d$ bounded.

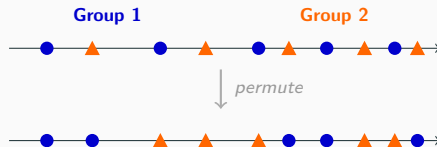
WoW:

$$\begin{cases} \mathcal{O}\left(\frac{\log \log n}{\log n} + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & d = 1 \\ \mathcal{O}\left(\frac{\log \log n}{\log n} + \frac{\log n}{\sqrt{n}} + \frac{\log m}{\sqrt{m}}\right), & d = 2 \\ \mathcal{O}\left(\frac{\log \log n}{\log n} + \frac{1}{n^{1/d}} + \frac{1}{m^{1/d}}\right), & d > 2. \end{cases}$$

HIPM:

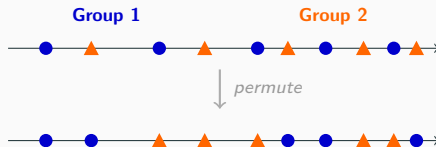
$$\begin{cases} \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & d = 1 \\ \mathcal{O}\left(\frac{\log n}{\sqrt{n}} + \frac{\log m}{\sqrt{m}}\right), & d = 2 \\ \mathcal{O}\left(\frac{1}{n^{1/d}} + \frac{1}{m^{1/d}}\right), & d > 2. \end{cases}$$

Practical Threshold via Permutation



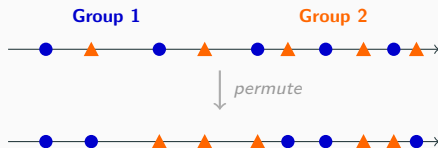
Practical Threshold via Permutation

- Observe $d(\hat{Q}_{n,m}^1, \hat{Q}_{n,m}^2)$. **Idea:** Estimate its $(1 - \theta)$ -quantile under H_0 .



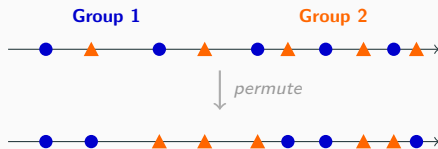
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- $\hat{P}_{1,m}^1, \dots, \hat{P}_{n,m}^1, \hat{P}_{1,m}^2, \dots, \hat{P}_{n,m}^2 \stackrel{\text{i.i.d}}{\sim}$ under H_0



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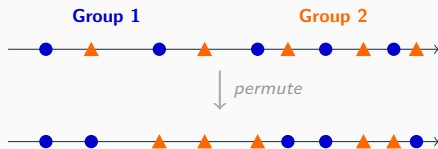
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- For $s = 1, \dots, S$: compute $d(\hat{Q}_{n,m}^{1,(s)}, \hat{Q}_{n,m}^{2,(s)})$



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- For $s = 1, \dots, S$: compute $d(\hat{Q}_{n,m}^{1,(s)}, \hat{Q}_{n,m}^{2,(s)})$
- **Threshold:** $(1 - \theta)$ -quantile of

$$d(\hat{Q}_{n,m}^{1,(1)}, \hat{Q}_{n,m}^{2,(1)}), \dots, d(\hat{Q}_{n,m}^{1,(S)}, \hat{Q}_{n,m}^{2,(S)})$$



Practical Threshold via Permutation

- **Test statistic:** $T_{n,m} = \sqrt{\frac{n}{2}} d_{\text{Lip}}(\hat{Q}_{n,m}^1, \hat{Q}_{n,m}^2)$
- **Permutation threshold:** for significance level $\theta \in (0, 1)$,

$$r(n, m, \theta) = \inf \left\{ t \in \mathbb{R} : \mathbb{P}(\tilde{T}_{n,m} > t \mid \mathbf{X}_{n,m}, \mathbf{Y}_{n,m}) \leq \theta \right\}.$$

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Theorem

Let $\mathbb{X} = [a, b]$ and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Then for all $\theta \in (0, 1)$

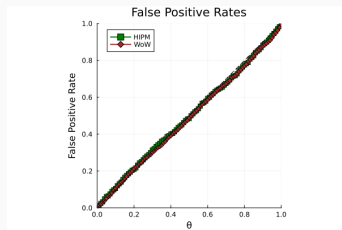
- if $Q^1 = Q^2$: $\lim_{n,m \rightarrow \infty} \mathbb{P}(T_{n,m} > r(n, m, \theta)) \leq \theta$.
- if $Q^1 \neq Q^2$: $\lim_{n,m \rightarrow \infty} \mathbb{P}(T_{n,m} > r(n, m, \theta)) = 1$.

WoW vs. HIPM: Type I error and Power

$$n = m = n_{\text{perm}} = 100.$$

Type I error

$$\text{DP}(\alpha = 1, P_0 = \text{Uniform}(0, 1))$$



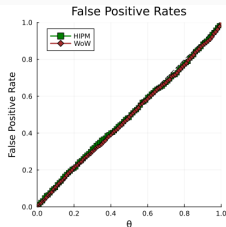
- Both methods **control Type I error**, including $\mathbb{X} = \mathbb{R}$ where theory does not guarantee it.

WoW vs. HIPM: Type I error and Power

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Type I error

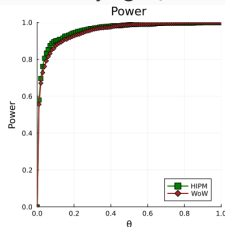
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Power: A.

$$\text{DP}(\alpha = 1, P_0 = \text{Beta}(a, b)),$$

varying P_0



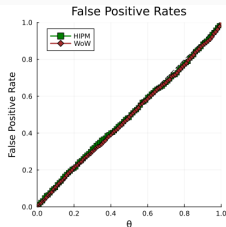
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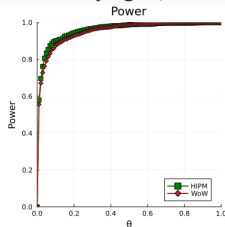
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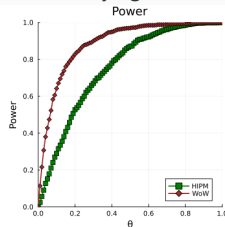
varying P_0



Power: B.

$$\text{DP}(\alpha = 1, P_0 = \mathcal{N}(0, 1)),$$

varying α



- Both methods **control Type I error**, including $\mathbb{X} = \mathbb{R}$ where theory does not guarantee it.
- **A:** HIPM slightly outperforms WoW.
- **B:** WoW outperforms HIPM.

- **Fréchet test** (Dubey and Müller 2019)
- **Energy test** (Székely and Rizzo 2004)
- **Distance-based test based on WoW**
- **Distance-based test based on HIPM**

Comparison with Existing Methods

Model: $\tilde{P}^i | \tilde{\mu} = \mathcal{N}(\tilde{\mu}, 1)$, $\tilde{\mu} \sim \mathcal{N}_T(\mu_{0i}, \sigma_{0i}^2, -10, 10)$.

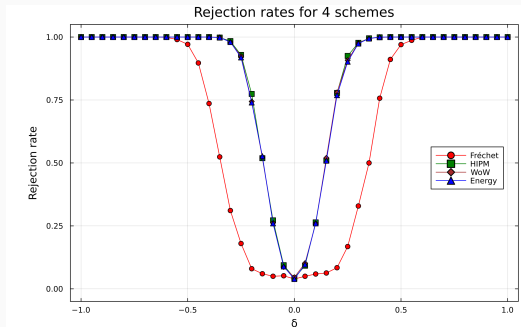
$n = 100$, $m = 200$, $\theta = 0.05$.

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Mean shift: $\sigma_{0i} = 0.5$, $\mu_{01} = 0$, $\mu_{02} = \delta$



- **Mean shift:** Energy, WoW, HIPM outperform Fréchet.

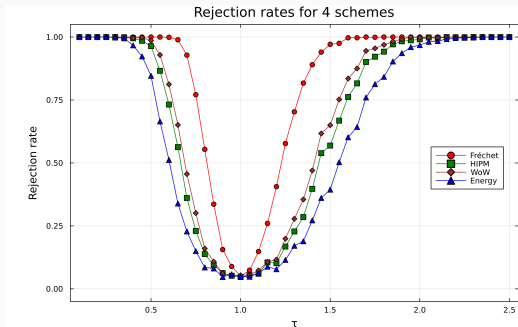
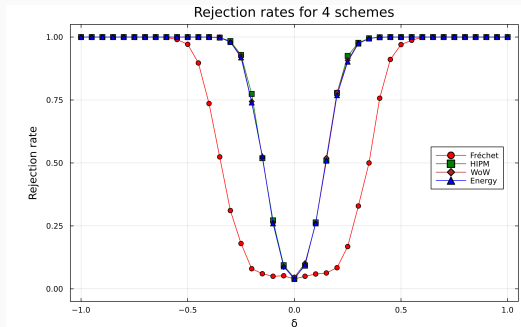
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Mean shift: $\sigma_{0i} = 0.5$, $\mu_{01} = 0$, $\mu_{02} = \delta$

Variance shift: $\mu_{0i} = 0$, $\sigma_{01} = 0.2$, $\sigma_{02} = 0.2\tau$



- **Mean shift:** Energy, WoW, HIPM outperform Fréchet.
- **Variance shift:** Fréchet > WoW > HIPM > Energy.
- All methods control Type I error ($\delta = 0$, $\tau = 1$).

Comparison with Existing Methods

- $Q^1 \neq Q^2$ but **same** Fréchet mean and variance.

$\tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, 1)$ in both cases, with

$$Q^1 : \tilde{\mu} \sim \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1,$$

$$Q^2 : \tilde{\mu} \sim \frac{1}{8}\delta_{-2} + \frac{3}{4}\delta_0 + \frac{1}{8}\delta_2.$$

Test family: (Q^1, M_λ) , $M_\lambda = \lambda Q^1 + (1-\lambda)Q^2$,
 $\lambda \in [0, 1]$.

$\lambda = 1$: Type I error; $\lambda < 1$: power.

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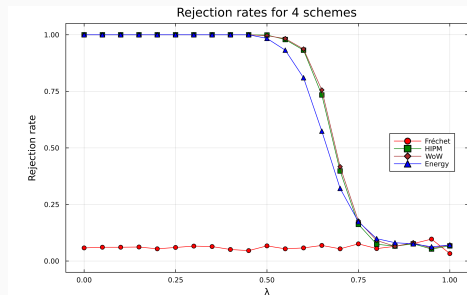
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$\lambda = 1$: Type I error; $\lambda < 1$: power.

- Fréchet test cannot detect $Q^1 \neq M_\lambda$.
- HIPM, WoW, and Energy correctly detect the discrepancy.



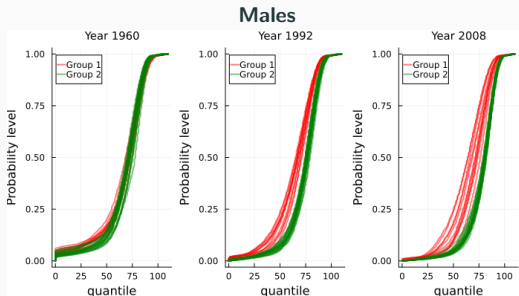
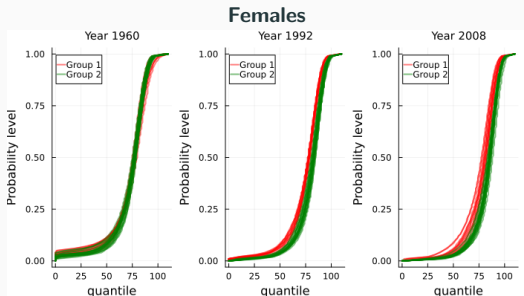
Application: Human Mortality Data

Data: Human Mortality Database, 1960–2010.

age-at-death distribution over ages $\{0, 1, \dots, 110\}$ per each country and year.

- Group 1: Belarus, Bulgaria, Czechia, Estonia, Hungary, Latvia, Lithuania, Poland, Russia, Slovakia, Ukraine
- Group 2: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Iceland, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Spain, Sweden, Switzerland, UK, USA.

Application: Human Mortality Data



Quantile plots of age-at-death distributions for years 1960, 1992, 2008.

- Years capture the Cold War era, post Soviet-collapse, and a more recent period.
- Differences are visually apparent in later periods.

Application: Human Mortality Data

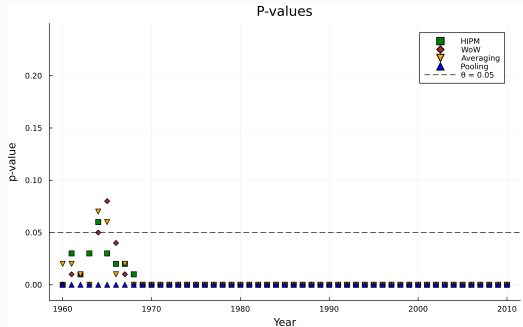
Methods:

- HIPM / WoW
- **Averaging:** $\bar{P}^k = \frac{1}{n_k} \sum_i \hat{P}_i^k$; $\mathcal{W}_1(\bar{P}^1, \bar{P}^2)$.
- **Pooling:** aggregate all deaths per group; permute at observation level.

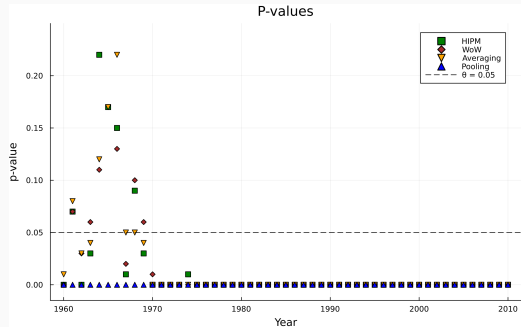
$$\theta = 0.05, n_{\text{perm}} = 100$$

Application: Human Mortality Data

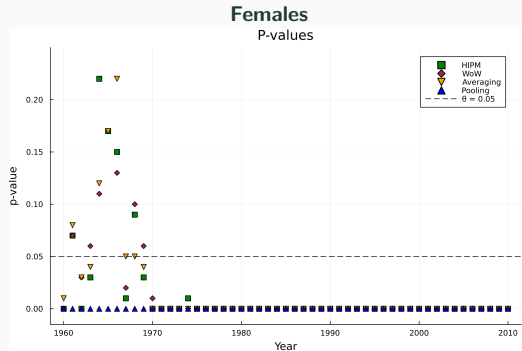
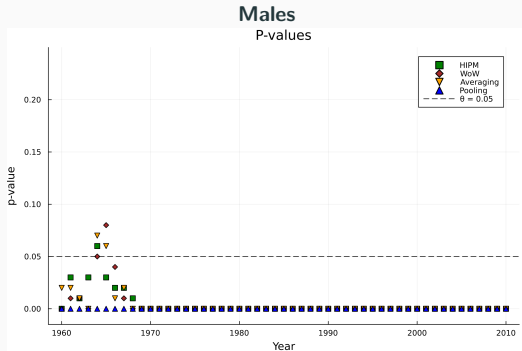
Males
P-values



Females
P-values

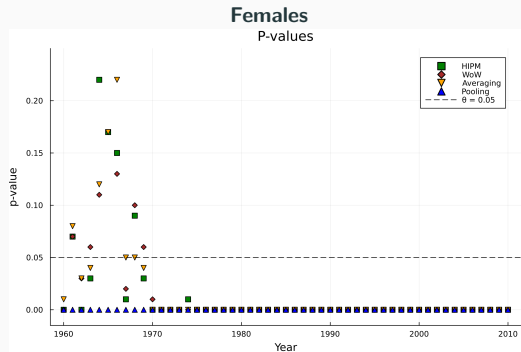
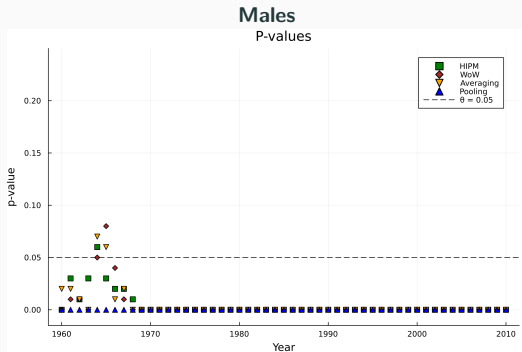


Application: Human Mortality Data



- HIPM, WoW, and Averaging show similar trends: significant differences after 1970s.

Application: Human Mortality Data



- HIPM, WoW, and Averaging show similar trends: significant differences after 1970s.
- **Pooling** fails to control Type I error in hierarchical sampling setting.

-  Baxter, Jonathan (2000). **“A Model of Inductive Bias Learning”**. In: *Journal of Artificial Intelligence Research* 12, pp. 149–198.
-  Catalano, Marta and Hugo Lavenant (2024). **“Hierarchical Integral Probability Metrics: A distance on random probability measures with low sample complexity”**. In: *Proceedings of the 41st International Conference on Machine Learning* 235, pp. 5841–5861.
-  Dubey, Paromita and Hans-Georg Müller (2019). **“Fréchet analysis of variance for random objects”**. In: *Biometrika* 106.4, pp. 803–821.
-  Gretton, Arthur et al. (2012). **“A Kernel Two-Sample Test”**. In: *Journal of Machine Learning Research* 13.25, pp. 723–773.
-  Haviv, Doron et al. (2024). **“Wasserstein flow matching: Generative modeling over families of distributions”**. In: *arXiv preprint arXiv:2411.00698*.

-  Hu, Xiaoyu and Zhenhua Lin (Apr. 2025). **“Two-sample distribution tests in high dimensions via max-sliced Wasserstein distance and bootstrapping”**. In: *Biometrika* 112.2, asaf001.
-  Szabó, Zoltán et al. (2016). **“Learning Theory for Distribution Regression”**. In: *Journal of Machine Learning Research* 17.152, pp. 1–40.
-  Székely, G. J. and M. L. Rizzo (2004). **“Testing for Equal Distributions in High Dimensions”**. In: *InterStat* 5, pp. 1249–1272.
-  Wang, Jie, Rui Gao, and Yao Xie (2022). **“Two-Sample Test with Kernel Projected Wasserstein Distance”**. In: *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics* 151, pp. 8022–8055.
-  Warren, Andrew et al. (2025). **“Principal Curves in Metric Spaces and the Space of Probability Measures”**. In: *arXiv preprint arXiv:2505.04168*.

Thank you for your attention!